Innate immunity — practice quiz

53 items. Answers and explanations are in answer-key.pdf.

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1. Type A — one best answer

A 19-year-old man with cystic fibrosis has repeated lower-airway bacterial infections. Sputum cultures grow bacteria that colonize the airway surface. Blood cultures stay negative and there is no evidence of tissue invasion. Which failure of innate defense best explains this pattern?

A Loss of tight junctions between airway epithelial cells

B Absent alpha-defensin production by Paneth cells

C Deficient histatin secretion by the salivary glands

D Impaired mucus clearance by ciliary action

E Lack of intraepithelial T lymphocytes with limited TCR diversity

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2. Type A — one best answer

A 35-year-old man is admitted with community-acquired pneumonia. On day 2 his temperature is 39.2 °C, his blood neutrophil count is raised, and his serum C-reactive protein (CRP) is many times above the upper reference limit. Which mechanism best explains the raised CRP?

A Plasma cells secrete CRP as an early, antigen-specific antibody

B Hepatocytes increase CRP synthesis in response to IL-6, IL-1 and TNF

C Neutrophils release preformed CRP stored in their granules

D Macrophage IL-10 drives hepatic CRP production

E The hypothalamus secretes CRP as part of the febrile response

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3. Type A — one best answer

A vaccine team tests a purified recombinant protein antigen in healthy volunteers. Alone, the protein induces only weak T cell–dependent antibody responses. When the team adds an aluminium salt, the responses become strong. What is the main reason for this effect?

A Aluminium increases the diversity of B-cell receptor genes

B Aluminium cross-links the T-cell receptor and so replaces signal 1

C Aluminium blocks IL-10 negative feedback in macrophages

D Aluminium induces memory in innate cells, so later responses are faster

E Aluminium activates innate responses that make DCs express costimulators and cytokines (signal 2)

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4. KPrim — mark each statement true / false

Innate recognition and the epithelial barrier. Which of the following statements is/are correct?

□ T □ F **A** Innate immune receptors are germline-encoded and distributed nonclonally: all cells of one lineage carry identical receptors.

□ T □ F **B** Lysozyme is more effective against Gram-negative than Gram-positive bacteria because it cleaves lipopolysaccharide.

□ T □ F **C** Paneth cells in the crypts of the small bowel are a major source of alpha-defensins.

□ T □ F **D** Human neutrophils produce functional theta-defensins as their main antimicrobial peptide.

5. KPrim — mark each statement true / false

Systemic effects of innate cytokines. Which of the following statements is/are correct?

□ T □ F **A** Endogenous pyrogens induce cyclooxygenase-2 to make prostaglandin E2, which acts on the hypothalamus; aspirin reduces fever by blocking prostaglandin synthesis.

□ T □ F **B** Acute-phase proteins have broad specificity for microbial patterns but no structural diversity.

□ T □ F **C** Anti-TNF antibodies have shown a clear survival benefit in clinical trials in patients with sepsis.

□ T □ F **D** TNF blockade can reactivate tuberculosis, which shows the role of TNF in keeping infection local.

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6. Type A — one best answer

A 46-year-old woman develops generalised urticaria and hypotension within minutes of an intravenous antibiotic given at induction of anaesthesia. She has never received this drug before. Specific IgE to the drug is negative. Which mast cell receptor most likely mediated this IgE-independent reaction?

A FcεRII (CD23)

B MRGPRX2

C FcγRIIB

D CD16 (FcγRIIIA)

E NKG2D

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7. Type A — one best answer

A 52-year-old man has had three episodes of unexplained anaphylaxis. Between episodes his baseline serum tryptase stays raised. A bone marrow biopsy shows clusters of mast cells with abnormal morphology. A mutation in which gene is most likely?

A GATA2

B CD18 (β2 integrin β chain)

C FcεRI α chain

D c-KIT

E LYST

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8. Type A — one best answer

In a laboratory assay, a tumour cell line that has lost all surface MHC class I is mixed with blood lymphocytes from a healthy donor who was never exposed to the tumour. CD56-positive, CD3-negative cells lyse the tumour cells within hours. No antibody is present. Which mechanism best explains the killing?

A Loss of inhibitory signals from KIRs and CD94/NKG2A ('missing self')

B T-cell receptor recognition of tumour peptide on MHC class I

C ADCC through CD16 binding to IgG on the tumour cells

D Engagement of the activating receptor CD94/NKG2A by HLA-E

E Release of IL-12 that directly lyses the tumour cells

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9. Type A — one best answer

A 16-year-old girl has had recurrent severe herpesvirus infections and extensive warts since childhood. Her blood shows almost no NK cells, and her monocyte, dendritic cell and B-cell counts are also low. Her mother has a similar history. Which is the most likely diagnosis?

A Chédiak-Higashi syndrome

B CD16 (FcγRIIIA) deficiency

C GATA2 deficiency

D Leukocyte adhesion deficiency type 1

E Chronic granulomatous disease

10. Type A — one best answer

A 27-year-old woman with allergic asthma has airway eosinophilia and mucus hypersecretion within hours of allergen exposure. Biopsy studies suggest that the first IL-5 and IL-13 come from tissue-resident cells that lack a T-cell receptor and that respond to IL-33, IL-25 and TSLP from the epithelium. Which cell is this?

A ILC1

B ILC3

C NK cell

D Th2 cell

E ILC2

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11. Type A — one best answer

A 38-year-old man with severe allergic asthma produces thick sputum. Microscopy shows many eosinophils and Charcot-Leyden crystals. Which eosinophil protein forms these crystals?

A Major basic protein

B Galectin-10

C Eosinophil peroxidase

D Tryptase

E Eosinophil cationic protein

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12. KPrim — mark each statement true / false

Mast cells, basophils and eosinophils in allergic disease. Which of the following statements is/are correct?

□ T □ F **A** Basophils express high levels of FcεRI and can be triggered when antigen binds IgE on their surface.

□ T □ F **B** Eosinophils are efficiently activated by allergen cross-linking of IgE bound to their FcεRI.

□ T □ F **C** Mature mast cells circulate in the blood and migrate to tissues during allergic reactions.

□ T □ F **D** Tryptase is present in all human mast cells, whereas chymase is present only in some mast cell subsets.

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13. KPrim — mark each statement true / false

Dendritic cells and NK cells. Which of the following statements is/are correct?

□ T □ F **A** Plasmacytoid dendritic cells express high levels of TLR7 and TLR9 and are the main producers of type I interferon.

□ T □ F **B** CD16 on NK cells binds IgE and mediates antibody-dependent cell-mediated cytotoxicity.

□ T □ F **C** The inhibitory receptor CD94/NKG2A recognises HLA-E, which displays peptides from other MHC class I molecules.

□ T □ F **D** Follicular dendritic cells are bone marrow-derived cells that present protein antigens to naive T cells.

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14. Type A — one best answer

A 2-year-old boy has his second episode of invasive pneumococcal infection (pneumonia with bacteraemia). His parents report that he hardly had a fever during either episode, and the local signs of inflammation were mild. He has had several common viral infections without complications. Which defect best explains this picture?

A Loss-of-function mutation in TLR3

B Mutation in the IL-12 receptor β1 chain

C Autosomal recessive IRAK4 deficiency

D Gain-of-function mutation in NLRP3

E Homozygous mutation in UNC93B

15. Type A — one best answer

A 6-year-old girl is admitted with her second episode of herpes simplex virus-1 encephalitis. She has no history of severe bacterial, fungal or other viral infections. Immunoglobulins and lymphocyte subsets are normal. A defect in which pathway is most likely?

A MyD88–IRAK4 signalling to NF-κB

B TLR5 recognition of bacterial flagellin

C NOD2 recognition of muramyl dipeptide

D Lectin pathway initiation by MBL

E Endosomal TLR3–TRIF signalling to type I IFN

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16. Type A — one best answer

A 34-year-old man has had attacks since childhood of fever, urticarial rash, joint pain and conjunctivitis. The attacks start a few hours after cold exposure. His father has the same attacks. Which mechanism best explains his disease?

A Gain-of-function NLRP3 mutation with excess inflammasome-generated IL-1β

B MEFV mutation with dysregulated pyrin inflammasome

C TNFRSF1A mutation affecting TNF receptor I

D Gain-of-function NOD2 mutation with excess NF-κB

E Partial loss of mevalonate kinase activity

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17. Type A — one best answer

A 58-year-old man has a third attack of acute monoarthritis of the first toe. Joint aspirate shows monosodium urate crystals. He does not respond to conventional anti-inflammatory drugs, and the rheumatologist proposes IL-1 blockade. Which innate mechanism explains why IL-1 blockade is effective?

A Urate crystals bind TLR4 and signal through MyD88 alone

B Cytosolic DNA from crystals activates cGAS–STING

C NOD2 in Paneth cells recognizes urate and activates RIP2

D Phagocytosed crystals damage lysosomes and activate NLRP3–caspase-1

E Cytosolic urate activates caspase-4 and caspase-5 directly

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18. Type A — one best answer

A 45-year-old woman with gram-negative urosepsis has a plasma CRP that rises steeply over 48 hours. Which group of cytokines, produced after TLR4 recognition of LPS, mainly drives this hepatic CRP synthesis?

A IL-12 and IL-18

B IL-6, IL-1 and TNF

C IFN-α and IFN-β

D IL-33 released as an alarmin

E CXCL8 and CCL2

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19. Type A — one best answer

A 5-month-old girl has recurrent fever, skin lesions due to cutaneous vasculopathy, and tachypnoea. Chest CT shows interstitial lung disease. Infection screening is negative. Which mechanism is most likely?

A Gain-of-function NLRP3 with excess IL-1β

B Loss-of-function TLR3 with low type I IFN

C Loss-of-function MyD88 with absent NF-κB signalling

D Gain-of-function NOD2 with excess NF-κB signalling

E Gain-of-function STING with excess type I IFN

20. KPrim — mark each statement true / false

Toll-like receptors. Which of the following statements is/are correct?

□ T □ F **A** TLR9 sits in endosomes and recognizes unmethylated CpG DNA.

□ T □ F **B** For LPS recognition, the accessory protein MD2 binds lipid A and the complex engages TLR4.

□ T □ F **C** TLR5 sits in endosomes and recognizes viral single-stranded RNA.

□ T □ F **D** TLR3 depends on MyD88 to induce IFN-β.

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21. KPrim — mark each statement true / false

NOD-like receptors and inflammasomes. Which of the following statements is/are correct?

□ T □ F **A** Caspase-1 cleaves gasdermin D, and the N-terminal fragment forms pores in the plasma membrane.

□ T □ F **B** Neutrophils are the main cell type that dies by pyroptosis.

□ T □ F **C** NOD2 recognizes muramyl dipeptide, and some NOD2 polymorphisms increase the risk of Crohn's disease.

□ T □ F **D** In humans, cytosolic LPS can activate caspase-4 or caspase-5 and cause pyroptosis.

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22. KPrim — mark each statement true / false

Soluble PRRs and cytosolic nucleic-acid sensors. Which of the following statements is/are correct?

□ T □ F **A** CRP binds phosphorylcholine and activates the classical complement pathway by binding C1q.

□ T □ F **B** Ficolins recognize carbohydrates through a C-type lectin domain, like MBL.

□ T □ F **C** RIG-I recognizes viral RNA with a 5′-triphosphate, which capped host mRNA lacks.

□ T □ F **D** MAVS is an endoplasmic reticulum adaptor that binds cGAMP.

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23. Type A — one best answer

An 8-month-old boy has recurrent bacterial skin infections and a wound that heals poorly. The infected sites show no pus. His blood leukocyte count is high. Flow cytometry shows very low CD11a, CD11b and CD11c on his neutrophils. Which gene is most likely mutated?

A GDP-fucose transporter

B KINDLIN-3

C CD18 (β2 integrin chain)

D LYST

E gp91 subunit of cytochrome b558

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24. Type A — one best answer

A 2-year-old girl has recurrent bacterial infections and a high blood leukocyte count. She also has severe mucosal bleeding that needed blood transfusions. Her platelet count is normal. Her skin, hair and eye pigmentation are normal. Neutrophil CD18 expression by flow cytometry is normal. Which mechanism best explains her disease?

A Absent expression of β2 integrins on leukocytes

B Absence of sialyl Lewis X on leukocytes

C Failure of NADPH oxidase to make superoxide

D Defective intracellular trafficking of lysosomes

E Defective inside-out activation of integrins

25. Type A — one best answer

A 4-year-old boy has a Staphylococcus aureus liver abscess. Last year he had invasive pulmonary aspergillosis. A lymph node biopsy shows granulomas. His maternal uncle died in childhood of a severe infection. Which molecular defect is most likely?

A Mutation in the gp91 subunit of cytochrome b558

B Mutation in the p47 cytosolic oxidase subunit

C Myeloperoxidase deficiency

D Mutation in the CD18 gene

E Mutation in IFN-γ receptor 1

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26. Type A — one best answer

A 35-year-old woman has occasional bacterial infections that respond well to treatment. Laboratory studies show that her neutrophils produce normal amounts of superoxide and hydrogen peroxide. However, her neutrophils cannot convert hydrogen peroxide into hypochlorous acid. Which enzyme is deficient?

A Superoxide dismutase

B Myeloperoxidase

C NADPH oxidase (gp91 subunit)

D Inducible nitric oxide synthase

E Neutrophil elastase

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27. Type A — one best answer

A 3-year-old boy has recurrent pyogenic infections, silvery hair and partial oculocutaneous albinism. His blood film shows giant granules in neutrophils and lymphocytes. He now has fever, hepatosplenomegaly and hemophagocytosis. Which protein is most likely defective?

A GDP-fucose transporter

B KINDLIN-3

C Neutrophil elastase (ELA2)

D LYST (lysosomal trafficking regulator)

E HAX1

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28. Type A — one best answer

A 6-year-old boy has episodes of fever and bacterial infection about every 3 weeks. Serial blood counts show neutrophil numbers that swing from near normal to almost undetectable on a roughly 21-day cycle. Which gene is most likely mutated?

A HAX1

B CD18

C ELA2 (neutrophil elastase)

D LYST

E IL-12Rβ1

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29. Type A — one best answer

A 1-year-old girl has recurrent bacterial infections, a high blood leukocyte count, severe developmental delay and the Bombay blood group phenotype. Which step of neutrophil recruitment is primarily blocked?

A Selectin-mediated rolling on endothelium

B Chemokine-induced integrin activation

C Integrin binding to ICAM-1 for firm arrest

D Paracellular transmigration via CD31

E Phagosome fusion with lysosomes

30. KPrim — mark each statement true / false

Regarding leukocyte recruitment from blood into infected tissue, which of the following statements is/are correct?

□ T □ F **A** IL-1 and TNF induce E-selectin on endothelial cells within 1 to 2 hours.

□ T □ F **B** LFA-1 on neutrophils binds mainly VCAM-1 on endothelium.

□ T □ F **C** Chemokine receptor signals raise integrin affinity by inside-out signalling.

□ T □ F **D** Neutrophils are recruited mainly through CCR2 binding to CCL2.

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31. KPrim — mark each statement true / false

Regarding microbial killing by neutrophils, which of the following statements is/are correct?

□ T □ F **A** Phagocyte oxidase reduces molecular oxygen to superoxide, with NADPH as cofactor.

□ T □ F **B** Myeloperoxidase converts superoxide into hydrogen peroxide.

□ T □ F **C** Catalase made by Staphylococcus destroys hydrogen peroxide, which helps explain this organism's importance in CGD.

□ T □ F **D** NET formation requires citrullination of histones by PAD4.

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32. KPrim — mark each statement true / false

Regarding Mendelian susceptibility to mycobacterial disease (MSMD), which of the following statements is/are correct?

□ T □ F **A** Live BCG vaccine can cause disseminated infection in affected patients.

□ T □ F **B** IL-12p40 deficiency removes both IL-12 and IL-23 activity.

□ T □ F **C** Total absence of the IFN-γ receptor causes exaggerated granuloma formation.

□ T □ F **D** The typical severe infections are invasive Aspergillus and catalase-positive staphylococci.

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33. Type A — one best answer

A 19-year-old male student has his second episode of meningococcal meningitis in two years. He has had no other serious infections. Complement C3 and C4 levels are normal. A defect in which complement function best explains this pattern?

A Opsonization of bacteria by C3b and iC3b

B Mast cell activation by the anaphylatoxins C3a and C5a

C Osmotic lysis by the membrane attack complex (C5b-9)

D Transport of immune complexes by CR1 on erythrocytes

E B-cell co-stimulation by C3d binding to CR2 (CD21)

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34. Type A — one best answer

A 45-year-old woman develops flushing, hypotension and wheeze during massive complement activation. Specific IgE to all tested triggers is negative. Which complement fragment is the most potent direct inducer of mast cell degranulation in this setting?

A C3b

B C4a

C C3a

D C5b-9

E C5a

35. Type A — one best answer

A 32-year-old man has episodes of intravascular hemolysis. Flow cytometry shows that his erythrocytes lack GPI-anchored membrane proteins. Which pair of complement-control functions is missing on his red cells?

A Dissociation of C3 convertases (DAF) and blocking of C9 binding to C5b-8 (CD59)

B Dissociation of C1r2s2 from C1q (C1-INH) and inhibition of MASP2

C Displacement of C2 from C4b in plasma (C4BP) and factor I cofactor activity

D Binding of S protein (vitronectin) to soluble C5b-7 complexes

E Factor H binding to sialic acid and displacement of Bb from C3b

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36. Type A — one best answer

A 4-year-old child has microangiopathic hemolytic anemia, thrombocytopenia and acute kidney injury. Genetic testing shows a heterozygous mutation in the factor H gene. Which normal function of factor H is most likely reduced?

A Direct proteolytic cleavage of C3b into iC3b without any cofactor

B Inhibition of C1r, C1s and MASP2 by covalent binding

C Blocking of C9 polymerization on the C5b-8 complex

D Binding of C3b on host cells to displace Bb and to act as factor I cofactor

E Stabilization of the C3bBb convertase on microbial surfaces

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37. Type A — one best answer

A 6-year-old boy has lobar pneumonia caused by Streptococcus pneumoniae. His serum C-reactive protein (CRP) is very high. CRP bound to the pneumococcal surface activates complement by binding which molecule?

A Mannose-binding lectin (MBL)

B C1q

C Factor B

D Properdin

E L-ficolin

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38. KPrim — mark each statement true / false

Complement convertases. Which of the following statements is/are correct?

□ T □ F **A** The classical and lectin pathways form the same C3 convertase, C4b2a.

□ T □ F **B** Factor D circulates as an inactive zymogen and is activated by C3b.

□ T □ F **C** The alternative pathway C5 convertase contains one Bb and two C3b molecules.

□ T □ F **D** C5b binds covalently to the target membrane through a thioester bond.

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39. KPrim — mark each statement true / false

Complement receptors and regulators. Which of the following statements is/are correct?

□ T □ F **A** C1-INH inhibits C1r, C1s and MASP2, and it also inhibits kallikrein and factor XII.

□ T □ F **B** CR2 (CD21) on B cells binds C3d and forms a co-receptor complex with CD19 and CD81.

□ T □ F **C** CD59 protects host cells by dissociating the C3 convertase C3bBb.

□ T □ F **D** Factor H regulates both pathways because it displaces C2a from C4b2a.

40. Type A — one best answer

A 19-year-old man has his second episode of meningococcal meningitis in 3 years. His maternal uncle died of meningococcal sepsis at age 20. His mother and sisters are healthy. Which inherited complement deficiency best explains this pattern?

A C2 deficiency

B C8 deficiency

C Properdin deficiency

D MBL deficiency

E C1q deficiency

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41. Type A — one best answer

A 24-year-old woman has a photosensitive rash, arthritis and positive antinuclear antibodies. She also had two episodes of bacterial pneumonia. Genetic testing finds a homozygous deficiency of a single complement component. This component deficiency is the most common in humans. Which component is deficient?

A C2

B C3

C C6

D Factor I

E Properdin

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42. Type A — one best answer

A 7-year-old boy has had recurrent pneumonia and one episode of bacteraemia with pyogenic bacteria. Serum C3 is markedly low. Genetic testing shows a homozygous factor I mutation; the C3 gene is normal. What best explains the low C3?

A Reduced hepatic synthesis of C3

B An autoantibody that destroys C3

C Loss of GPI-anchored regulators on blood cells

D Uncontrolled fluid-phase C3 convertase that consumes C3

E Failure of C1-INH to block C1r and C1s

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43. Type A — one best answer

A 16-year-old girl has recurrent painful abdominal attacks and swelling of the hands since age 11. Her father has similar episodes. Antihistamines did not help. She has no wheals. Which laboratory profile is most likely, if she has the most common form of hereditary angioedema?

A C1-INH function normal, C1-INH protein normal, C4 normal

B C1-INH function low, C1-INH protein normal or high, C4 low

C C1-INH normal, C3 low, CH50 low

D C1-INH function low, C1-INH protein low, C1q low

E C1-INH function low, C1-INH protein low, C4 low

44. Type A — one best answer

A 63-year-old man has had four episodes of facial and tongue swelling in 6 months. He has no family history of angioedema and no wheals. He takes no ACE inhibitor. C1-INH function, C1-INH protein and C4 are all low on two occasions. Which test is the most useful next step?

A Serum tryptase

B Serum C1q

C SERPING1 gene sequencing

D C3 and CH50

E F12 gene analysis

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45. Type A — one best answer

A 29-year-old woman has recurrent lip and facial swellings lasting 2-3 days, without wheals. They became more frequent after she started a combined oral contraceptive. Her father has similar swellings. High-dose antihistamines did not help. C1-INH concentration and C1-INH function are normal on repeated testing. What is the most likely diagnosis?

A HAE-2 (dysfunctional C1-INH)

B Acquired angioedema with C1-INH deficiency

C Hereditary angioedema with normal C1-INH

D ACE inhibitor-induced angioedema

E Mast cell-mediated angioedema of chronic spontaneous urticaria

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46. Type A — one best answer

A 35-year-old man with known HAE-1 comes to the emergency department 1 hour after onset of tongue swelling and a hoarse voice. He has no stridor yet. According to the 2017 WAO/EAACI guideline, what is the best management?

A Intramuscular adrenaline, IV antihistamine and IV glucocorticoid, then observe

B High-dose oral tranexamic acid and admission for observation

C Start danazol and review the response after 24 hours

D Give C1-INH concentrate or icatibant at once and prepare early airway intervention

E Observe without drugs because most attacks resolve without treatment

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47. Type A — one best answer

A 31-year-old woman with HAE-1 is 16 weeks pregnant. Her attacks have increased to three per month, with two abdominal attacks. She asks for long-term prophylaxis. According to the 2017 WAO/EAACI guideline, which option is preferred?

A Danazol at the lowest effective dose

B Icatibant twice weekly

C Ecallantide every 3 days

D Tranexamic acid as first choice

E Plasma-derived C1-INH concentrate

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48. KPrim — mark each statement true / false

Complement deficiencies and their phenotypes. Which of the following statements is/are correct?

□ T □ F **A** Deficiency of C5, C6, C7, C8 or C9 predisposes mainly to disseminated Neisseria infection.

□ T □ F **B** Properdin deficiency is inherited as an autosomal dominant trait.

□ T □ F **C** More than half of patients with C1q deficiency develop SLE.

□ T □ F **D** MBL deficiency typically causes severe lifelong pyogenic infections similar to C3 deficiency.

49. KPrim — mark each statement true / false

Management of HAE-1/2 according to the WAO/EAACI guideline, 2017 revision. Which of the following statements is/are correct?

□ T □ F **A** Icatibant is a competitive antagonist of the bradykinin B2 receptor.

□ T □ F **B** C1-INH concentrate is the preferred short-term prophylaxis before a dental extraction.

□ T □ F **C** Attenuated androgens are the recommended first-line long-term prophylaxis.

□ T □ F **D** Every patient should have on-demand medication for two attacks and carry it at all times.

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50. KPrim — mark each statement true / false

Diseases caused by defective complement regulators. Which of the following statements is/are correct?

□ T □ F **A** PNH results from a germline PIG-A mutation inherited from a parent.

□ T □ F **B** Mutations in factor H, MCP or factor I predispose to atypical haemolytic uraemic syndrome.

□ T □ F **C** Factor I deficiency causes secondary C3 depletion and pyogenic infections.

□ T □ F **D** Factor H mainly regulates the classical pathway C3 convertase.

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51. SCT

A 38-year-old woman has had recurrent swellings of the lips and hands for several years and two episodes of severe abdominal pain. She has no wheals. Daily antihistamines did not prevent the swellings. You consider the causes of recurrent angioedema. (Provisional panel credits.)

Hypothesis: HAE-1/2 (C1-INH deficiency or dysfunction)
New info: C1-INH function is low and C4 is low, confirmed on a repeat sample.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

Hypothesis: Acquired angioedema with low C1-INH (AAE-C1-INH)
New info: C1q is normal.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

Hypothesis: HAE with normal C1-INH (HAE nC1-INH)
New info: C1-INH function, C1-INH protein and C4 are normal on two occasions, and her mother has the same swellings.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

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52. SCT

A 6-year-old boy has had recurrent bacterial infections since infancy, including skin abscesses and two pneumonias. Growth is normal. You consider a phagocyte defect or a complement deficiency. (Provisional panel credits.)

Hypothesis: Chronic granulomatous disease (CGD)
New info: He now has a liver abscess with Staphylococcus aureus and a lung infection with Aspergillus.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

Hypothesis: Terminal complement (C5-C9) deficiency
New info: He has had two episodes of disseminated Neisseria meningitidis infection.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

Hypothesis: Leukocyte adhesion deficiency type 1 (LAD-1)
New info: The skin abscesses contained abundant pus.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

53. SCT

A 9-year-old girl has had recurrent episodes of fever with an urticaria-like rash, joint pain and red eyes since early childhood. Episodes often follow exposure to cold. Her mother had similar episodes as a child. (Provisional panel credits.)

Hypothesis: Cryopyrin-associated periodic syndrome (CAPS, NLRP3 gain of function)
New info: Her mother has progressive hearing loss.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

Hypothesis: Familial Mediterranean fever (FMF)
New info: Genetic testing finds a gain-of-function NLRP3 mutation.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2

Hypothesis: CAPS (NLRP3 gain of function)
New info: The attacks respond dramatically to anakinra, an IL-1 receptor antagonist.
Effect on hypothesis: □ −2 □ −1 □ 0 □ +1 □ +2